

CREDIT OPPORTUNITY MEMORANDUM

Credit Suisse Group AG (CS) | Financials — Global Systemically Important Bank (G-SIB)

Prepared using the Corporate Credit Risk Early-Warning System | Confidential — For Internal Discussion Purposes

Deal Context

This memorandum presents a retrospective credit risk assessment of Credit Suisse Group AG (“CS” or “the Bank”), evaluated through a proprietary composite early-warning model that combines bank-specific accounting ratios with a market-implied Merton distance-to-default measure. The purpose of this exercise is to demonstrate that CS's credit deterioration was observable and quantifiable well in advance of its emergency acquisition by UBS in March 2023 — the model's composite risk score crossed our defined distress threshold in 2022Q2, three quarters ahead of resolution, with the market-based signal (distance-to-default) providing the earliest confirmation.

Executive Summary

- **Credit Suisse's composite credit risk score rose steadily from 57.1% (2022Q1) to 67.3%** in its final full trading quarter (2023Q1), crossing our 60% distress threshold in 2022Q2 — approximately three quarters before the UBS-brokered emergency rescue announced in March 2023.
- **Distance-to-default (Merton model) flagged the earliest and sharpest signal:** DD fell from 3.05 (2022Q1, baseline solvency) to 1.74 (2022Q2, implied PD 4.12%), well ahead of the accounting ratios' deterioration.
- **Peer comparison against JPMorgan Chase and Goldman Sachs** — both of which remained range-bound between 40–50% throughout the same period — confirms the deterioration was CS-specific, not sector-wide.
- **Recommendation / Rating: HIGH RISK — DISTRESS ZONE** (Composite Score 67.3%, pre-resolution). Consistent with the actual outcome of external resolution via UBS acquisition.

Business & Industry Overview

Credit Suisse was a global systemically important bank headquartered in Zurich, operating across wealth management, investment banking, and Swiss universal banking. Heading into 2022, the Bank was already contending with a series of legacy risk-management and litigation issues (including the Archegos and Greensill episodes) that had eroded market confidence ahead of the period covered by this analysis.

The broader banking sector entered a period of acute stress in 2022–23 as aggressive central bank rate hikes compressed the value of fixed-income asset holdings and tightened funding conditions industry-wide, culminating in the March 2023 failures of Silicon Valley Bank and Signature Bank in the U.S. CS's collapse occurred within this same window but was driven primarily by idiosyncratic, CS-specific deposit flight rather than the asset-liability mismatch that drove the U.S. regional bank failures — a distinction this analysis supports through the deposit and funding data below.

Credit Analysis

Composite Risk Score Trajectory

The model's composite score, derived from a logistic regression fit against sector-appropriate bank signals (CET1 ratio, loan-to-deposit ratio, non-performing loan ratio, net interest margin) and distance-to-default, tracked a consistent upward trajectory through five quarters of active trading data:

Quarter	CS Composite Score	JPM Composite Score	GS Composite Score
2022Q1	57.1%	45.0%	45.1%
2022Q2	61.4% (crosses 60% threshold)	44.9%	49.6%
2022Q3	63.7%	43.6%	48.6%

2022Q4	65.2%	40.2%	48.4%
2023Q1	67.3% (final active quarter — UBS rescue)	44.9%	49.5%

Note: CS was delisted following the March 2023 UBS acquisition. Post-resolution quarters in the underlying dataset carry an imputed distress-floor value and are excluded from this table as they do not represent live market data.

Distance-to-Default: The Earliest Signal

The market-implied distance-to-default measure moved ahead of the accounting-based composite score, providing the earliest confirmation of deteriorating solvency:

Quarter	Equity Volatility (σ_E)	Distance-to-Default (DD)	Implied PD (%)	Status / Milestone
2022Q1	33.8%	3.05	0.11%	Baseline solvency
2022Q2	55.8%	1.74	4.12%	Early warning triggered
2022Q3	76.5%	1.00	15.84%	Accelerated distress
2022Q4	98.2%	0.55	29.11%	Severe deterioration
2023Q1	135.4%	-0.08	53.09%	UBS rescue quarter

Deposit & Funding Stress

The most acute stress signal in the dataset is the deposit base itself. Total deposits fell from \$390B to \$180B — a **54% decline** — over five quarters, pushing the loan-to-deposit ratio from 76.9% to 144.4% over the same period:

Quarter	Total Deposits	CET1 Capital	Loan-to-Deposit Ratio
2022Q1	\$390B	\$41.0B	76.9%
2022Q2	\$360B	\$39.5B	81.9%
2022Q3	\$310B	\$38.0B	93.5%
2022Q4	\$230B	\$36.0B	121.7%
2023Q1	\$180B	\$31.0B	144.4%

Capital Structure

As a G-SIB, CS's capital structure included a substantial layer of Additional Tier 1 (AT1) contingent convertible bonds sitting subordinate to senior unsecured debt and senior to common equity in ordinary circumstances. A defining feature of the actual UBS resolution was that Swiss regulator FINMA exercised a full write-down of approximately CHF 16 billion in AT1 notes to zero as part of the rescue — while equity holders retained a residual (if severely diluted) recovery through UBS's stock-and-cash offer. This inverted the conventional capital structure waterfall (equity subordinate to AT1) and is a widely cited case study in contingent-capital risk. This analysis's declining CET1 trend (from \$41.0B to \$31.0B over the period, against broadly stable risk-weighted assets) is consistent with the capital erosion that ultimately triggered this outcome.

Red Flags

- **CET1 capital declined from \$41.0B to \$31.0B (–24%)** over five quarters while risk-weighted assets remained broadly stable, signaling erosion of the Bank's core solvency buffer rather than balance-sheet shrinkage alone.
- **Loan-to-deposit ratio rose from 76.9% to 144.4%**, moving from a conservative funding profile to one indicating significant reliance on non-deposit funding — a classic precursor to a liquidity-driven crisis.
- **Equity volatility more than tripled (33.8% → 135.4%)**, directly compressing distance-to-default from a safe 3.05 to a negative reading (-0.08) by the final active quarter.

- **The composite score's deterioration was monotonic** across all five observed quarters with no quarter of improvement or stabilization — a pattern that, in hindsight, offered no false-recovery signal to mask the underlying trend.

Peer Benchmarking

Metric (2023Q1 / Latest Active)	Credit Suisse	JPMorgan Chase	Goldman Sachs
Composite Risk Score	67.3%	44.9%	49.5%
Distance-to-Default	-0.08	6.70	5.30
Implied Probability of Default	53.09%	< 0.0001%	< 0.0001%
Loan-to-Deposit Ratio (2022Q1→Latest)	76.9% → 144.4%	Stable, ~53–54%	N/A / stable

Downside Sensitivity

Had deposit outflows continued at the 2022Q4→2023Q1 run-rate (approximately \$50B/quarter) for two additional quarters without external intervention, the loan-to-deposit ratio would have breached approximately 200%, and distance-to-default — already negative at -0.08 — would imply a probability of default approaching certainty on the model's logistic scale, underscoring why external resolution occurred when it did rather than the Bank stabilizing organically.

Risk Factors

- **Model risk:** the composite score is derived from a 14-company sample (7 failures), a small dataset by conventional statistical standards; bank-sector coefficients in particular are based on only 4 companies and should be treated as directionally indicative rather than precisely calibrated.
- **The deposit flight captured here reflects reported quarterly balances**, not daily/weekly outflow velocity; the actual acute run in March 2023 occurred on a timescale faster than this quarterly framework can resolve.
- **No standalone funding-cost (e.g., interest expense/liabilities) metric was incorporated;** loan-to-deposit ratio serves as the model's sole funding-stress proxy.

Recommendation / Rating

Based on the composite score of **67.3% (DISTRESS ZONE, >60% threshold)**, a negative and rapidly falling distance-to-default, and accelerating deposit attrition, this analysis would have assigned Credit Suisse a **HIGH RISK / DISTRESSED** rating as of 2023Q1 — consistent with, and in fact anticipating, the actual outcome of the Bank's emergency acquisition by UBS later that quarter. The earliest actionable signal in this framework (distance-to-default crossing into warning territory in 2022Q2) preceded the resolution event by approximately three quarters, which this analysis holds out as the model's central, defensible finding.